Lecture 2:

Electromechanical Switching

Introduction

- The Electromechanical switching systems are a combination of mechanical and electrical switching types.
- The electrical circuits and the mechanical relays are deployed in them.
- The Electromechanical switching systems are further classified
  - Step-by-step (Strowger switching)
  - Crossbar

Strowger switching system

- Strowger switching system was the first automatic switching system developed by Almon B.Strowger in 1889.
- Annoyed at the amount of business he was losing everyday.
- Strowger decided to make a switching system that would replace the human operator.
- The switch developed by him is named after him.
- Functionally, the system is classified as step-by-step switching system since the connections are established in a **step-by-step process**.

- In a manual exchange. The subscriber needs to communicate with the operator and a common language becomes an important factor.
- In multilingual areas this aspect may pose problems.
- On the other hand, the Operation of an automatic exchange is language independent.
- A greater degree of privacy is obtained in automatic exchanges as no operator is normally involved in setting up and monitoring a call
- Establishment and release of calls are faster in automatic exchanges

Dialing

how a call is made and how dialing is done without the help of an operator?

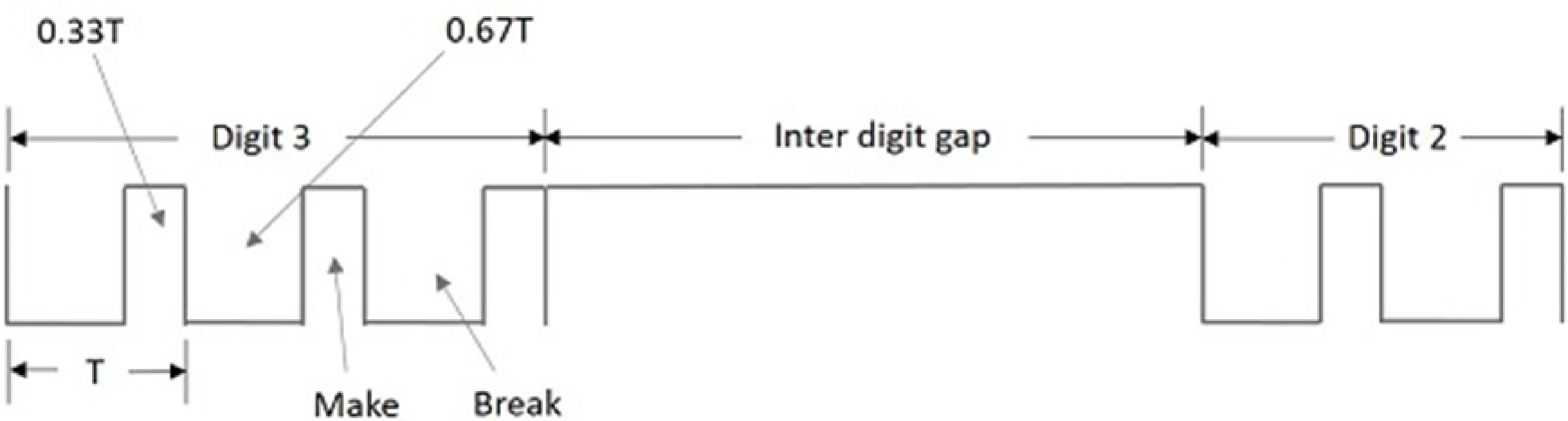
- Unlike in Manual Switching system, an automatic switching system requires a formal numbering plan or addressing scheme to identify the subscribers.
- Numbering plan is where a number identifies a subscriber, is more widely used than the addressing scheme in which a subscriber is identified by the alpha numerical strings
- There needs to be a mechanism to transmit the identity of the called subscriber to the exchange.

- This mechanism should be present in the telephone set, in order to connect the call automatically to the required subscriber.
- The methods prevalent for this purpose are **Pulse Dialing** and **Multi Frequency Dialing**.
- Of them, the Pulse dialing is the most commonly used form of dialing till date.

Pulse Dialing

- As the name implies, the digits that are used to identify the subscribers are represented by a train of pulses.
- The number of pulses in a train is equal to the digit value it represents except in the case of zero, which is represented by 10 pulses.
- Successive digits in a number are represented by a series of pulse trains.
- These pulses have equal number of time intervals and the number of pulses produced will be according to the number dialed

- Two successive trains are distinguished from one another by a pause in between them, known as the **Inter-digit gap.**
- The pulses are generated by alternately breaking and making the loop circuit between the subscriber and the exchange.
- An example pulse train is shown in the following figure.



- The above figure shows the pulsating pattern.
- The pulse rate is usually 10 pulses per second with a 10 percent of tolerance.
- The gap between the digits, which is called the Inter-digit gap is at least 200ms

Drawbacks of Pulse Dialing

- The pulse dialing technique is where there is making and breaking of the subscriber loops.
- This might disturb and affect the performance of speaker, microphone and bell contained in the telephone.
- In addition, the dialing timings should not affect the timing of the pulse train as this will lead to the dialing of a wrong number

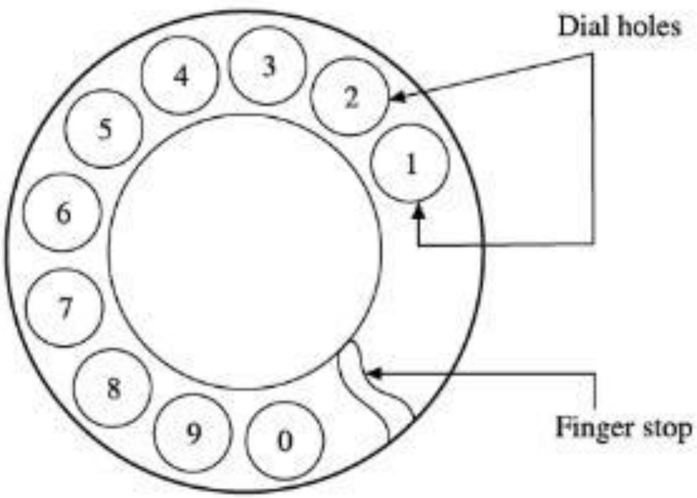
Rotary Dial Telephone

- The Rotary Dial Telephone came into existence to solve the problems prevailing then.
- The microphone and the loudspeaker are combined and placed in the receiver set.
- The set has a finger plate the arrangement of which makes the dialing time appropriate.
- The below figure shows how a rotary dial looks like

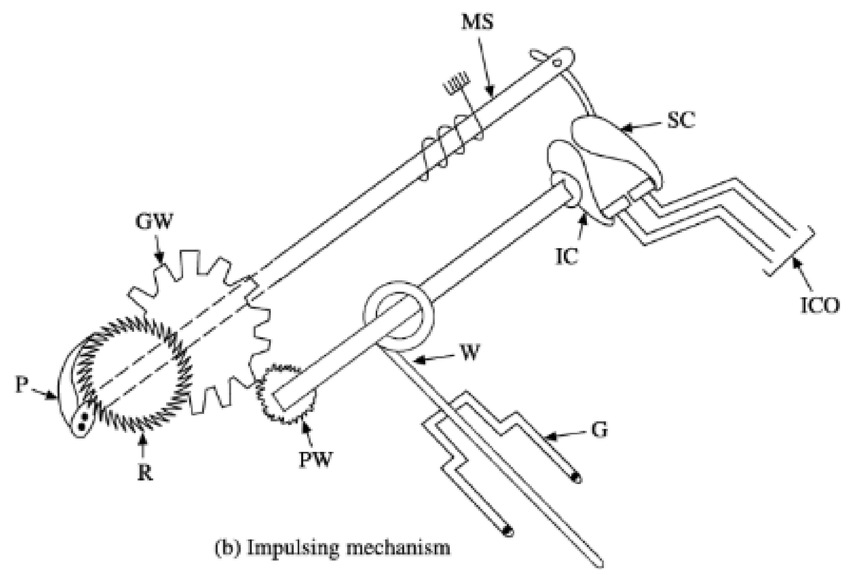
Rotary Dial Telephone

A rotary dial telephone uses the following for implementing pulse dialling:

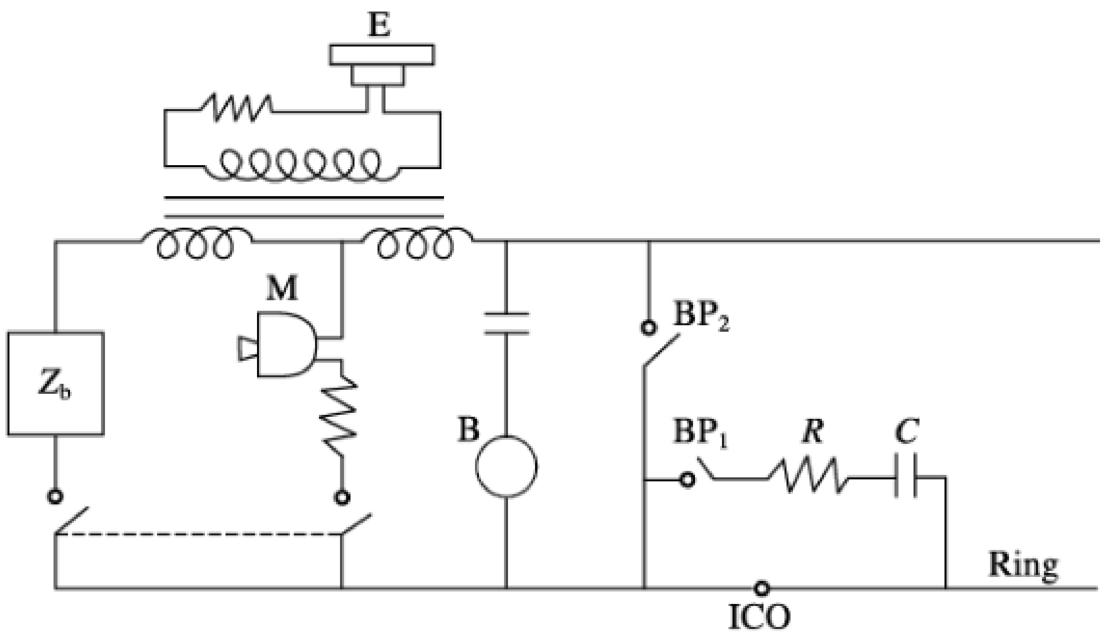
- ✓ Finger plate
- ✓ Shaft, gear and pinion wheel
- ✓ Pawl and ratchet mechanism
- ✓ Impulsing cam and suppressor cam or a trigger mechanism
- ✓ Impulsing contact
- ✓ Centrifugal governor and worm gear
- ✓ Transmitter. receiver and bell by-pass circuits.



(a) Finger plate arrangement



(b) Impulsing mechanism



B = bell BP_1, BP_2 = by-pass switches ICO = impulsing contact

Signaling Tones

- As the manual exchanges were replaced, the operator who used to communicate the calling subscribers regarding the situation of the called subscribers, needed to be replaced with different tones indicating different situations

Signaling Tones

A number of signaling functions are involved in establishing, maintaining and releasing a telephone conversation.

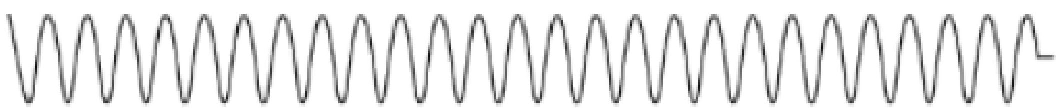
- Dial Tone
- Ring Tone
- Busy Tone
- Number Unobtainable Tone
- Routing Tone or Call-in-Progress Tone

Signaling Tones

- Respond to the calling subscriber that system is ready to receive the identification of the called party.
- Inform the calling subscriber that the call is being established.
- Ring the bell of the called party.
- Inform the calling subscriber, if the called party is busy.
- Inform the calling subscriber, if the called party line is unobtainable for some reason.

Signaling Tones

- Most often, the dial tone is sent out by the exchange even before the handset is brought near the ear.

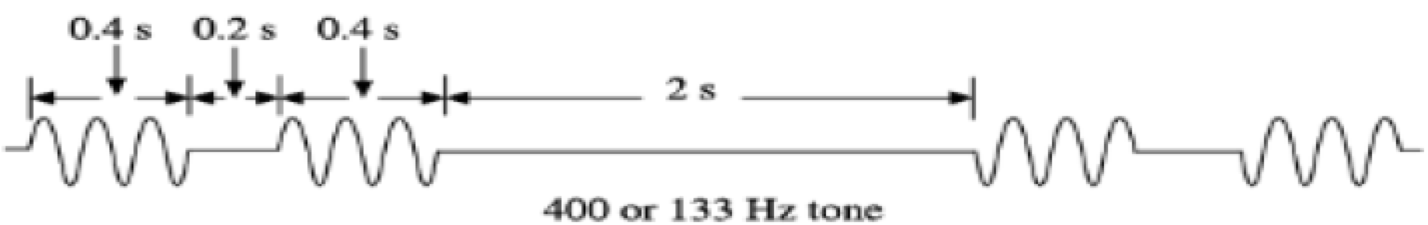


33 or 50 or 400 Hz continuous

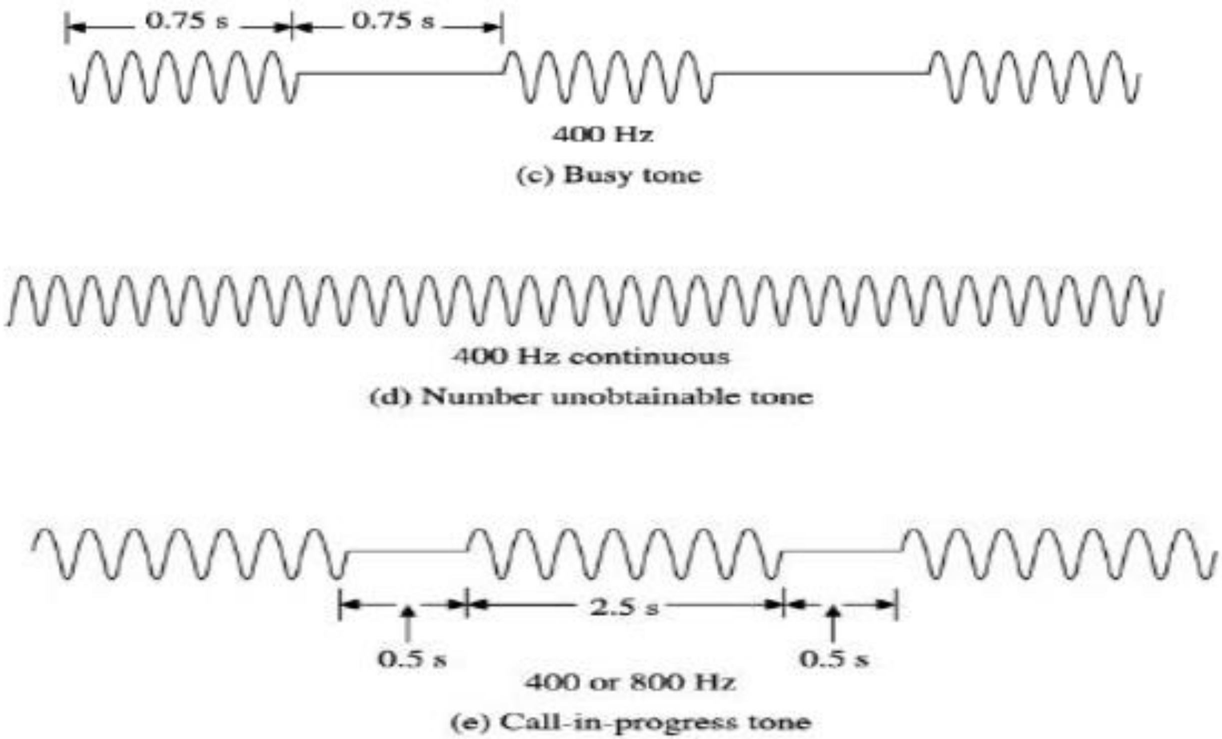
(a) Dial tone

Signaling Tones

- When the called party line is obtained, the exchange control equipment sends out the ringing current to the telephone set of the called party.
- This ringing current has the familiar double-ring pattern, simultaneously.
- The control equipment sends out a ringing tone to the calling subscriber.



(b) Ringing tone


0.75 s
0.75 s
400 Hz
(c) Busy tone
400 Hz continuous
(d) Number unobtainable tone
0.5 s
2.5 s
0.5 s
400 or 800 Hz
(e) Call-in-progress tone

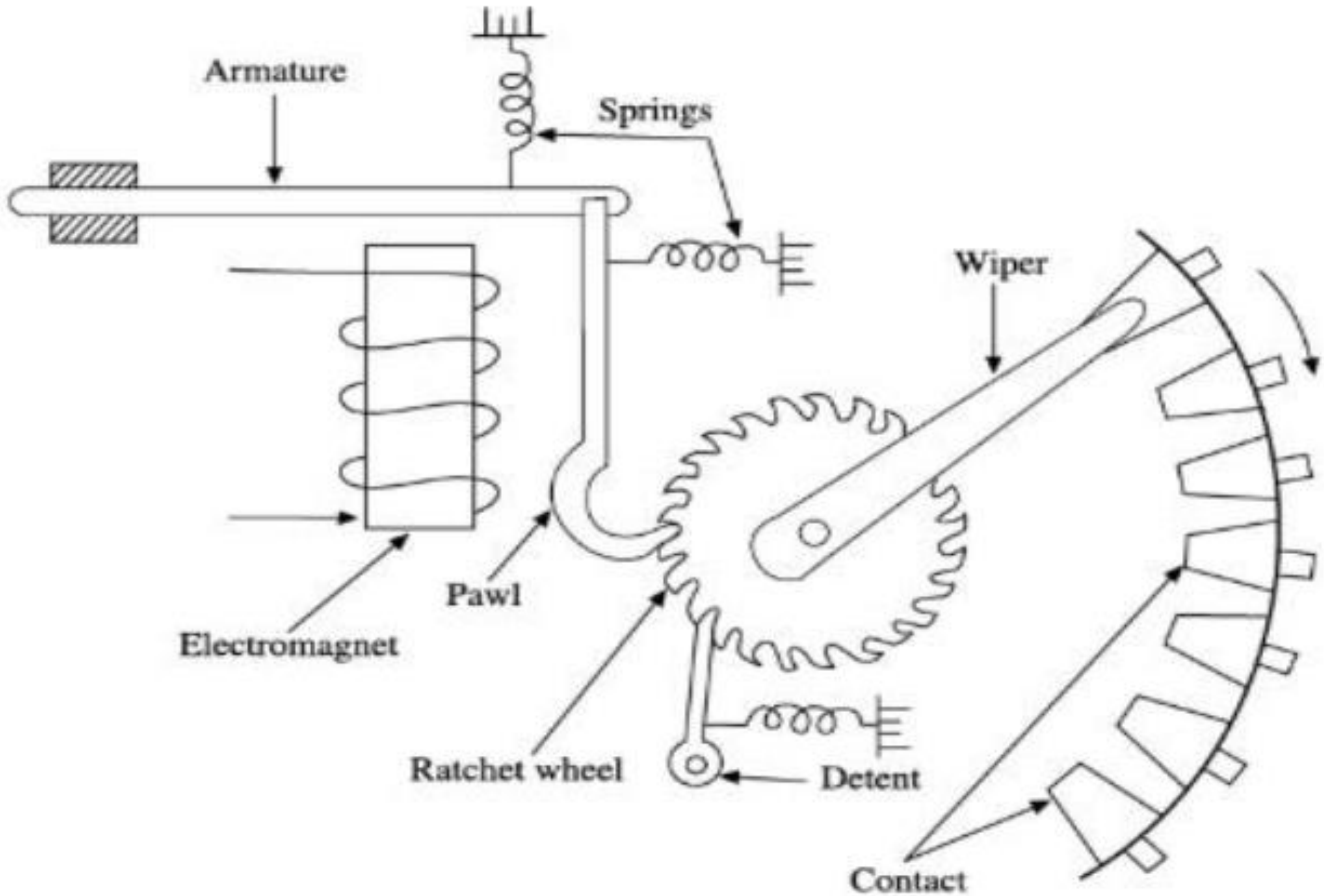
STROWGER SWITCHING COMPONENTS

In Strowger exchanges, two electromechanical devices form the basic building blocks for the switching system:

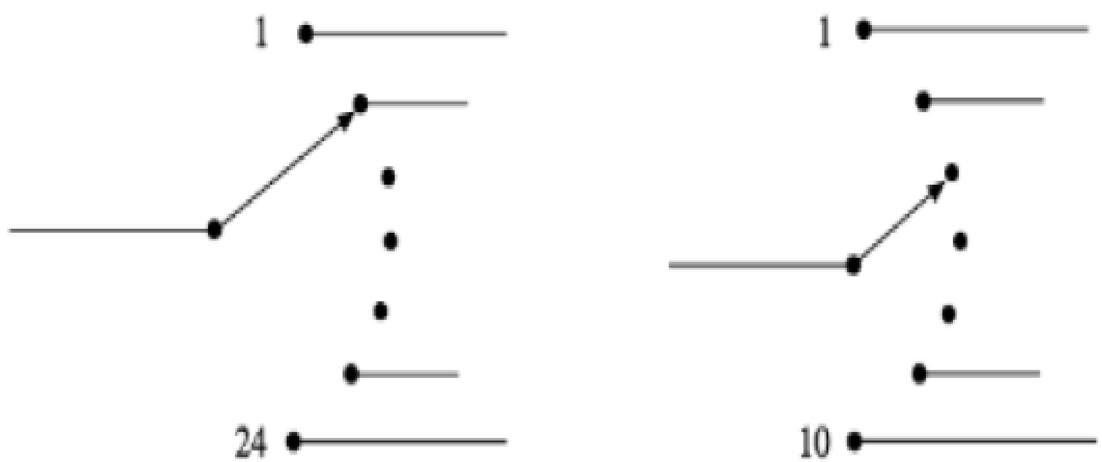
- Uni selector
- Two-motion selector

Uni selector

- These devices are called selectors as they enable selection of suitable switching paths for establishing connections amongst subscribers.
- **Uniselectors** have a mechanism to span across a bank of contacts that lie in a horizontal plane and to select a free contact that can be used to establish a connection
- **Two-motion selectors** have the ability to move both in the vertical and horizontal directions to select a free contact.
- Uniselectors and two-motion selectors are constructed using electro-mechanical rotary switches which in turn are built using among others, an electromagnet, an armature and a rachet wheel.



(a) Drive mechanism of a rotary switch



(b) Schematic representation of uniselectors

The proper functioning of a uniselector is dependent on a number of factors:

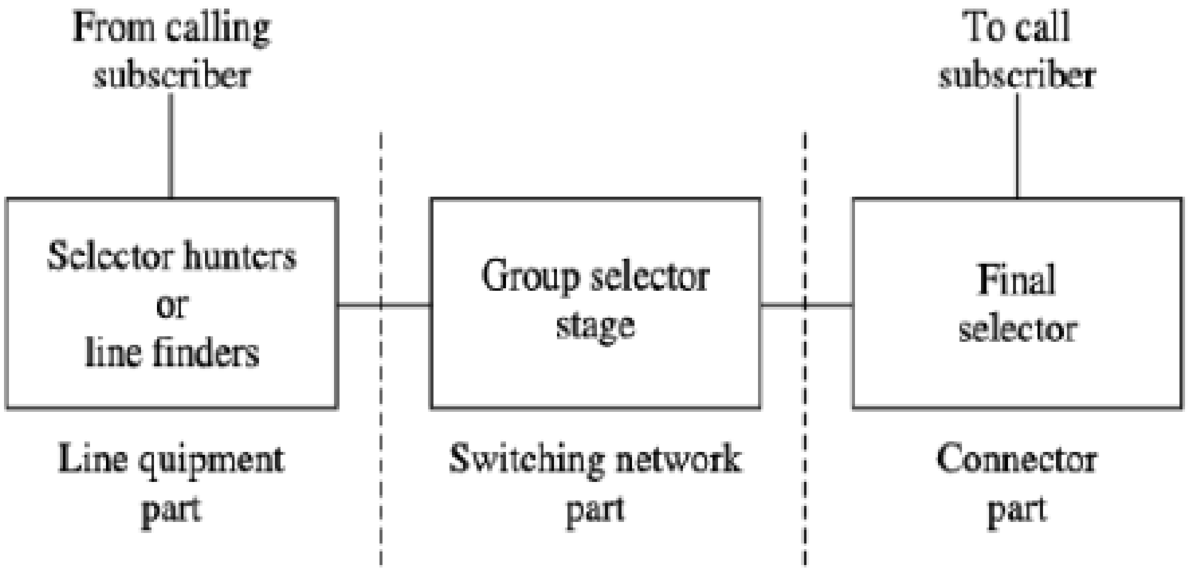
- Energizing current level
- Inertia of the moving system
- Friction between wipers and bank contacts
- Friction in drive assembly
- Tension in restoring springs
- Adjustment of interrupter contact.

Step by Step Switching

- A step-by-step switching system may be constructed using uniselectors or two-motion selectors or a combination of both. As already stated, the wiper contacts of these selectors move in direct response to dial pulses.

- They may also be made to move through interrupter mechanism which may be activated by other signals like off-hook from the subscriber telephone.

- The wiper steps forward by one contact at a time and moves by as many contacts (takes as many steps) as the number of dial pulses received or as required to satisfy certain signalling conditions.



Configuration of a step-by-step switching system.

Crossbar switching

- The Crossbar exchanges were developed during 1940s.
- They achieve full access and non-blocking capabilities with the Crossbar switches and common control equipment, used in the Crossbar exchanges.
- The active elements called Cross points are placed between the input and the output lines.
- In the common control switching systems, the separation between switching and control operations allows the usage of switching networks by a group of common control switches to establish many calls at the same time on a shared basis.

The Features of Crossbar Switches

- While processing a call, the common control system helps in the sharing of resources.

- The specific route functions of call processing are hardwired because of the Wire logic computers.

- The flexible system design helps in the appropriate ratio selection is allowed for a specific switch.

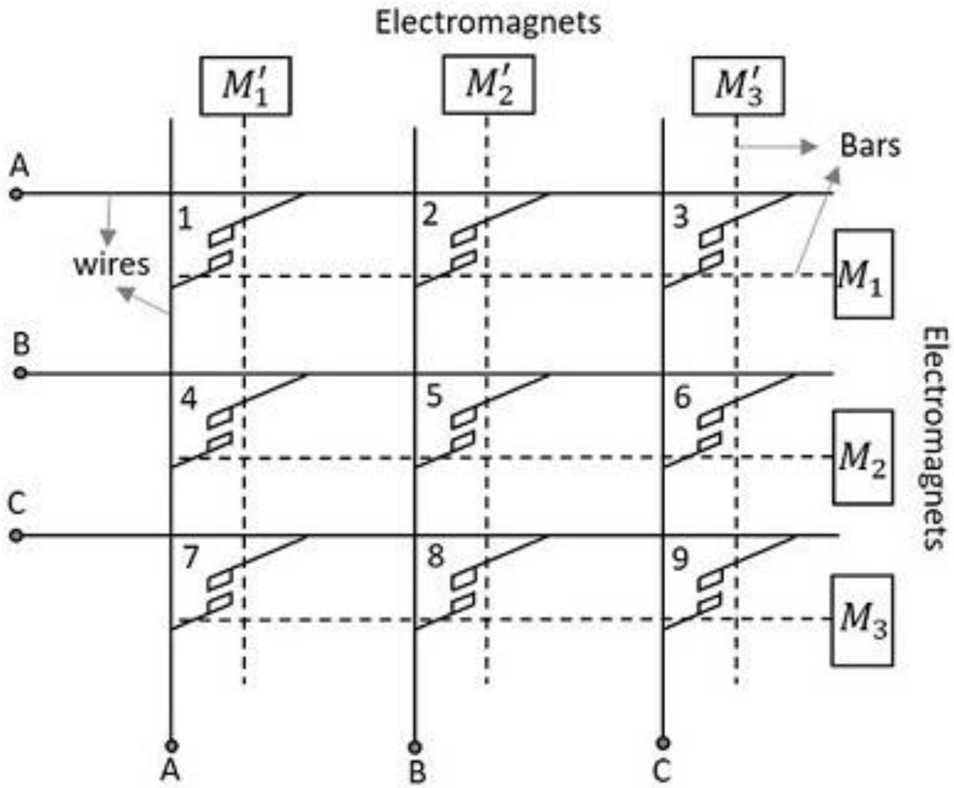
- Fewer moving parts ease the maintenance of Crossbar switching systems

Uses of Crossbar Switching

- The Crossbar switching system uses the common control networks which enable the switching network to perform event monitoring, call processing, charging, operation and maintenance.
- The common control also provides uniform numbering of subscribers in a multi-exchange area like big cities and routing of calls from one exchange to another using the same intermediate exchanges.

Crossbar Switching Matrix

- The Crossbar arrangement is a matrix which is formed by the M X N sets of contacts arranged as vertical and horizontal bars with contact points where they meet.
- They need nearly M + N number of activators to select one of the contacts.
- The Crossbar matrix arrangement is shown in the following figure.

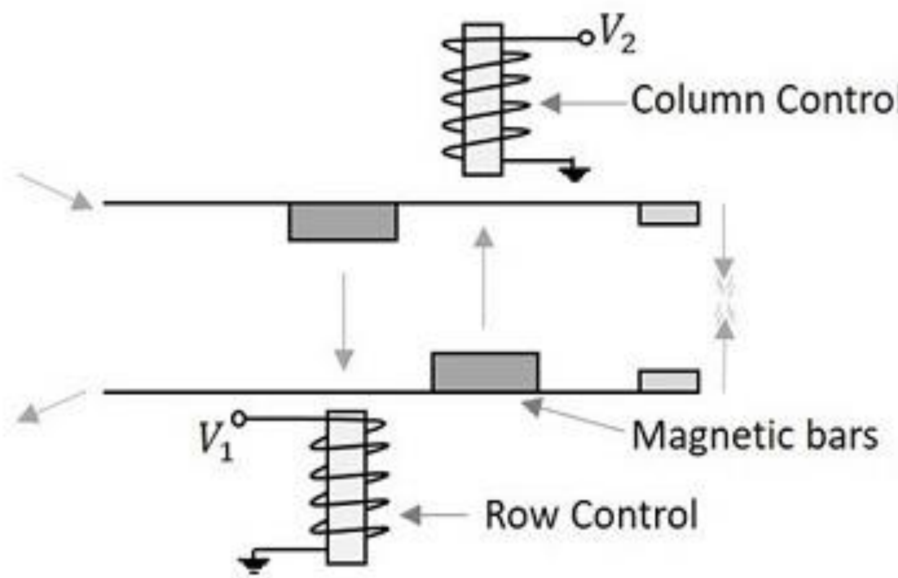


3 X 3 Crossbar Switching

- The Crossbar matrix contains an array of horizontal and vertical wires shown by solid lines in the following figure, which are both connected to initially separated contact points of switches.
- The horizontal and vertical bars shown in dotted lines in the above figure are mechanically connected to these contact points and attached to the electromagnets.

- The Crosspoints placed between the input and the output lines have electromagnets which when energized, close the contact of intersection of the two bars.
- This makes the two bars to come closer and hold on.
- The following figure will help you understand the contact made at the Crosspoints.

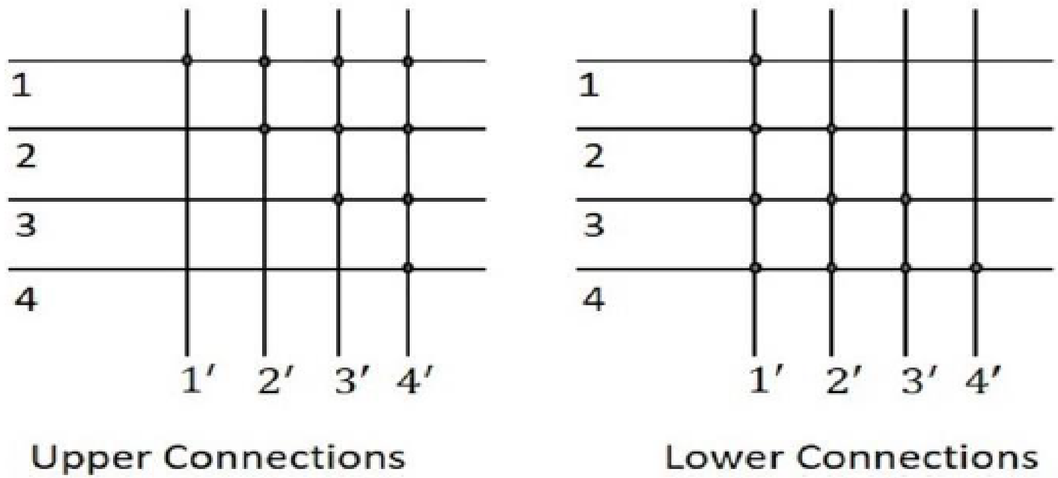
- Once energized, the electromagnets pull the small magnetic slabs present on the bars.
- The column control electromagnet pulls the magnet on the lower bar, while the row control electromagnet pulls the magnet on the upper bar



Diagonal Cross point Matrix

- In the matrix, as 1,2,3,4 indicate input lines and 1′,2′,3′,4′ indicate output lines of the same subscribers, if a connection has to be established between the 1st and the 2nd subscriber, then 1 and 2′ can be connected or 2 and 1′ can be connected using the Crosspoints.
- In the same way, when a connection has to be established between 3 and 4, then 3-4′ Crosspoint or 4-3′ Crosspoint can do the work.
- The following figure will help you understand how this works.

Diagonal Cross point Matrix



The diagonal points are also considered, the total number of Cross points will be

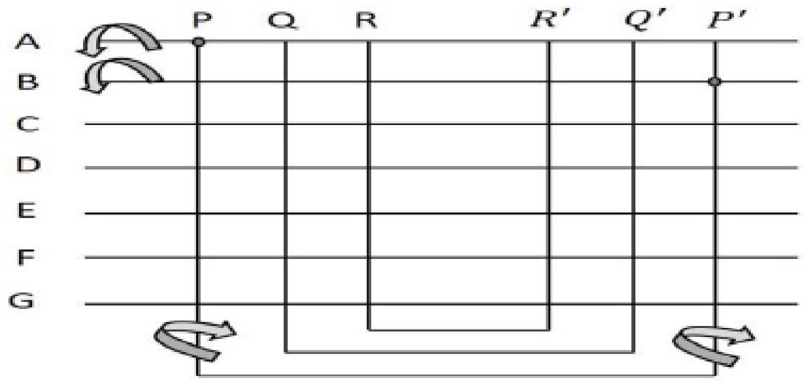
$$N\,(N+1)/2$$

If the diagonal points are not considered, then the total number of Cross points will be,

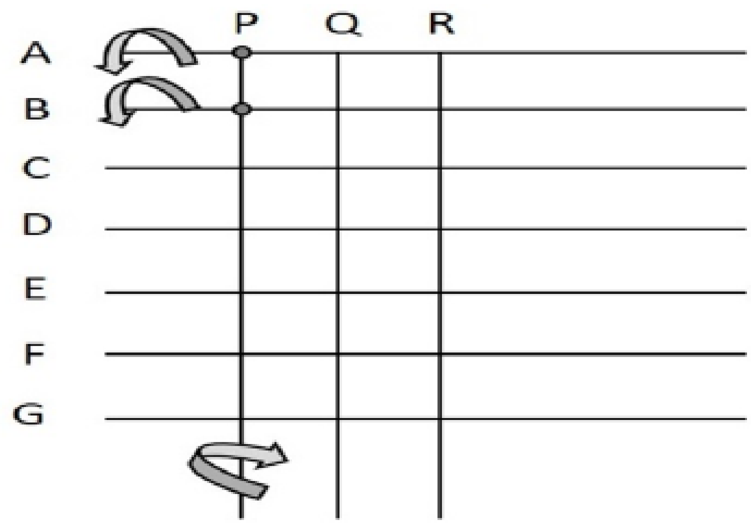
$$N\,(N-1)/2$$

Crossbar Switch Configurations

Blocking Crossbar Switches



- Hence, the steps associated with the establishment of connection follows a sequence:
- Energize horizontal bar A
- Energize free vertical bar P
- De-energize horizontal bar A
- Energize horizontal bar B
- Energize free vertical bar P' (associated with P)
- De-energize horizontal bar B



Hence, the establishment of connection follows a sequence:

- Energize horizontal bars A and B
- Energize free vertical bar P
- De-energize horizontal bars A and B

Thank You!